

The Two-Time Moving-Defect Problem for Dispersing Billiards

A go/no-go audit, dynamic graph currents, and a Fourier route

Working note

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Abstract

We isolate the estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of the scatterers. A tempting bridge from the Demers–Liverani projective cone to forward standard families fails: the moving-face coefficient uses a one-sided trace on a transversal terminal arc, whereas the cone controls stable-leaf integrals. We give a local counterexample to any inference of such a trace bound from the published cone seminorms. We prove a two-sided nuclear/recovery criterion which yields the desired product estimate, but emphasize that it is only a sufficient condition and is probably too strong for a graph delta. The intrinsic problem is instead two-parameter equidistribution of a centered graph current under $T^{-1} \times T$. A targeted literature audit points to an induced generalized-baker/Fourier-decay strategy: encode the collision SRB measure on a two-sided return scheme, prove Fourier decay for the induced slice by non-concentration or L^2 flattening, and lift the estimate through the exponential tower tail and the grazing blocks. An exact third-order calculation at a normal two-cycle between equal circular scatterers gives a nonzero Anosov cocycle, equal to -1 in a determinant-one adapted chart. This supplies a fixed open geometric QNL witness. A countable-branch Fourier theorem of Leclerc–Paukkonen–Sahlsten makes the previously proposed smooth finite-horseshoe exhaustion unnecessary: a uniformly contracting countable IFS with bounded distortion, a light-tailed Gibbs potential, and UNI already has QNL and power Fourier decay. We reduce its application here to a single-state full-branch billiard quotient and one finite two-word UNI calculation. The exact cocycle does not by itself identify those two words. We state the remaining falsifiable estimates and the quantitative margins needed for this program. No unconditional CM2 theorem is claimed here.

1 The exact target

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_X \|f\|_{C^1}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. The same estimate, uniformly for $s^{-1}(P_s - P_0)$, is required before a parameter limit can be exchanged with the two time sums.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_\Gamma, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u|_I\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C \varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

$\square$

Remark 3.3. *The theorem makes clear why a usual functional correlation estimate with error $O(\tau^m + \vartheta^n)$ is insufficient. CM2 requires a centered operator ideal estimate at the event time, not just mixing on either side. Wormell’s specialized two-time conditional-mixing conjecture is one way to encode precisely this product structure.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (6) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j(W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

4.1 Recovery windows and the actual summability target

For differentiability of the susceptibility, a uniform pointwise CM2 bound for every grazing block is stronger than necessary. The required conclusion is absolute summability of the complete block series. This distinction removes an artificial exponential loss in the grazing rank.

Proposition 4.1 (Recovery-window summability). *Let $B_b(m, n)$ be the scalar pairing produced by block b at past and future times m, n . Suppose that there are masses M_b , recovery times R_b^-, R_b^+ , constants $C < \infty$, and $0 < \tau, \vartheta < 1$ such that*

$$|B_b(m, n)| \leq C M_b \tau^{(m-R_b^-)_+} \vartheta^{(n-R_b^+)_+}, \quad m, n \geq 0. \quad (11)$$

Then

$$\sum_{b, m, n} |B_b(m, n)| \leq C' \sum_b M_b (1 + R_b^-)(1 + R_b^+). \quad (12)$$

In particular, if $M_b \leq C2^{-\alpha b}$ for some $\alpha > 0$ and $R_b^\pm \leq C_0 + C_1 b$, then the full threefold series converges. The same conclusion holds uniformly for parameter difference quotients if the constants and block masses are uniform.

The deterministic recovery times may be replaced by a countable refinement $B_b = \sum_j B_{b,j}$ with masses $M_{b,j}$ and times $R_{b,j}^\pm$, provided

$$\sum_j M_{b,j} (1 + R_{b,j}^-)(1 + R_{b,j}^+) \leq C M_b (1 + b)^2. \quad (13)$$

It is enough for the two-sided recovery tail, after its deterministic $O(b)$ entrance time, to be exponential.

Proof. For every $R \geq 0$ and $0 < q < 1$,

$$\sum_{k \geq 0} q^{(k-R)_+} \leq R + 1 + \frac{1}{1-q}.$$

Sum (11) first in m, n and then in b . Exponential block mass dominates the quadratic recovery window. Applying the same calculation to every refined piece gives (13). An exponential tail for $R_{b,j}^- + R_{b,j}^+ - O(b)$ has uniformly bounded polynomial moments, proving the last assertion. $\square$

Lemma 4.2 (Two marginal growth tails imply bi-recovery). *Let a block of total variation M_b be cut by the common refinement of its past and future recovery trees, with refined masses $M_{b,j}$ and recovery times $R_{b,j}^\pm$. Assume cutting preserves total variation and, for some $r_b \leq C_0 + C_1 b$, $0 < \rho < 1$,*

$$\sum_{j: R_{b,j}^\pm > r_b + t} M_{b,j} \leq C M_b \rho^t, \quad t \geq 0,$$

for each sign. Then (13) holds.

Proof. If $R_{b,j}^- + R_{b,j}^+ > 2r_b + t$, at least one marginal time is larger than $r_b + t/2$. The union bound therefore gives an exponential tail for the sum of the two recovery times under the normalized weights $M_{b,j}/M_b$. Its second moment is $O((1+r_b)^2)$. Since $(1+R^-)(1+R^+) \leq (1+R^-+R^+)^2$, the claim follows. $\square$

Proposition 4.3 (Growth-lemma reduction of proper bi-recovery). *Assume that every grouped physical face block admits a common finite initial refinement with the following properties.*

- (i) *Its past projection is a homogeneous stable-cone curve and, after at most r_0 one-sided terminal steps, its future projection is a homogeneous unstable-cone curve.*
- (ii) *In rescaled strip coordinates their initial lengths are bounded below by $c2^{-qb}$ for some fixed $q < \infty$.*
- (iii) *Cutting by either marginal growth tree and transporting the coarea weight preserve the sum of total variations up to a uniform constant.*

Then the usual forward growth lemma and its time reversal give the two marginal hypotheses of Lemma 4.2, with $r_b \leq C(1+b)$. Hence proper bi-recovery contributes only $C M_b(1+b)^2$ to the absolute susceptibility sum.

Proof. Uniform cone expansion enlarges a curve of length $c2^{-qb}$ to the reference length after at most $C(1+b)$ iterates. The growth lemma gives an exponential tail for descendants which are cut before reaching that length. Apply it to the future unstable projection and, by reversibility, to the past stable projection. Property (iii) permits a common refinement without changing the two marginal tail estimates except by a uniform factor. Lemma 4.2 completes the proof. $\square$

For a circular radial face, the equation $G_{j \rightarrow i} = 0$ identifies the past arc as a one-step preimage singularity, hence as a stable-cone singularity curve. Its one-sided terminal image lies on the grazing seam; after the fixed tangent-skip convention and one further clean collision, time reversal identifies the continuation as an unstable-cone singularity curve.

The square-root blown-up impact equation gives polynomial strip length, which is property (ii). The only apparently new part is therefore the uniform total-variation/distortion statement (iii). For the prescribed clean radial first defect it is already contained in the measure-level physical assembly, as the next corollary records.

Corollary 4.4 (Clean radial blocks satisfy the bi-recovery summation). *For the same-label grouped first radial face current in a fixed physical square-root atlas, assume the clean one-sided terminal continuation used to define the complete first defect. For prescribed compact regular observables, properties (i)–(iii) of Proposition 4.3 hold uniformly on a small table chart. Consequently, conditional only on the central shifted estimate (11), all high homogeneity blocks contribute an absolutely summable series. The conclusion is uniform for the assembled first difference quotients.*

Proof. The past cone and future cone statements are the standard singularity-curve geometry described above. The blown-up impact equation and fixed-atlas jet bounds give a polynomial lower bound for the rescaled component lengths, so (ii) holds for some finite q . For (iii), a common refinement only restricts the signed coarea measure to disjoint measurable pieces; the sum of their total variations is the original total variation. Push-forward by an injective homogeneous branch preserves total variation, while the standard bounded-distortion estimate restores the uniform log-Hölder density bound after recovery. The physical block estimate supplies the uniform initial mass, including for the assembled difference quotient. Apply Proposition 4.3 and then Proposition 4.1. $\square$

Thus the earlier warning after Proposition 4.3 concerns a general unresolved graph current. For the clean prescribed radial first defect, the measure-level argument above closes it. It does not address second shape defects, point atoms, or an arbitrary completed trace-domain seed.

For a general graph current the relevant geometric input is therefore a *proper bi-recovery* statement, not an estimate of a Fourier constant on the original grazing curve. For the clean first radial current this input is Corollary 4.4. A central two-time theorem then gives (11). The unrecovered rectangle $m < R_{b,j}^-$, $n < R_{b,j}^+$ is handled by the crude total-variation bound; the growth-lemma tail sums these windows through (13).

Consequently even the generic $M_b = O(2^{-b})$ assembly margin is sufficient once proper bi-recovery is proved; the stronger first-radial $O(2^{-(3-\epsilon_*)b})$ margin gives additional room but is no longer needed to absorb an exponential Fourier constant in b . This decouples the grazing rank from the return-rank truncation in Proposition 6.4.

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (14)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (15)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (15) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;

- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (16)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (16), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (16) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (16) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b\gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m}\gamma_b(t))f(T^n F_b\gamma_b(t))w_b(t) dt. \quad (17)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b\gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (18)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_*\nu_b}(\xi)| \leq C_b(1 + |\xi|)^{-\eta} \quad (19)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (20)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(19) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

The proposition deliberately isolates the missing theorem. Uniform (19) cannot be inferred from one-sided mixing. It must be proved from non-concentration and nonlinearity of the induced inverse branches and the face graph. The L^2 -flattening framework and nonlinear-pushforward theorems of Baker–Khalil–Sahlsten and Baker–Banaji are plausible tools for this step. Wormell proves the corresponding Fourier mechanism for classes of generalized baker maps, but not the present two-time billiard statement.

6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole (m, n) quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

Lemma 6.2 (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (21)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}, \quad (22)$$

with all four exponents positive. Choose $A > \beta_-/\alpha_+$ and $D > \beta_+/\alpha_-$. Suppose in addition that

$$|B_{m,n}| \leq C e^{-\gamma \max\{m,n\}}$$

whenever $D^{-1}m < n < Am$. Then there are $\tau, \vartheta \in (0, 1)$ such that $|B_{m,n}| \leq C' \tau^m \vartheta^n$ for all m, n .

Proof. If $n \geq Am$, (21) is bounded by $e^{-c_+(m+n)}$ for some $c_+ > 0$. If $m \geq Dn$, use (22). The remaining indices lie in the stated diagonal band, where $\max\{m, n\} \geq (m+n)/2$. Taking the minimum of the three positive exponents proves the claim. $\square$

For billiards, (21) is the expected output of signed standard-family coupling after cutting the past-weighted density into homogeneous pieces; the factor $e^{\beta_- m}$ records the growth of its regularity. Equation (22) follows by time reversal. Consequently the genuinely new Fourier/QNL theorem is needed only for $m \asymp n$.

This also corrects the naive ambient phase test. The graph $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$ is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable–unstable temporal-distance/holonomy function. A second-derivative QNL condition is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.3 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (23)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (23) follows. On the tangency face, $|((d, 0) - q_j) \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in (36) have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.4 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq C e^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq C e^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus the precise new input is a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails can then remove the truncation. What remains open is uniform control of the temporal-distance non-concentration constants as the truncation grows.

6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an η_{cut} fraction of its parent's conditional mass. For an arbitrary exhaustion by finite subshifts, η_{cut} may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank L . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (24)$$

with η_0 independent of x, L , for at least one QNL-separated child at each scale. If (24) and a uniform UNI constant $\kappa_0 > 0$ hold, Leclerc's tree argument gives a QNL exponent $\Gamma > 0$ independent of L ; only the multiplicative constant is allowed to grow like e^{cL} .

Rühr's pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (24) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

Lemma 6.5 (Uniform interval UNI implies power non-concentration). *Let ν_L be measures on an interval whose densities satisfy $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$, uniformly in L . Let X_L be a family of functions with the following scale-invariant UNI property: for every interval J of length $\sigma < \sigma_0$ and every polynomial P in a fixed finite-dimensional class, there is a subinterval $J' \subset J$ with $|J'| \geq \kappa|J|$ such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where $\kappa > 0$ is independent of J, P, L . Then for every fixed $\alpha > 0$ there are $C, \Gamma > 0$, independent of L , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

Proof. Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval J' . The density bounds imply that at least a fixed fraction $\eta_0 > 0$ of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to $|J|^{1+\alpha}$. The number of generations is $c_\alpha |\log |J|| + O(1)$, so the surviving mass is at most $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C |J|^\Gamma \nu_L(J)$. $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

6.5 A fixed two-branch UNI certificate

Bertozzi's 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let h_0, h_1 be two regular inverse return branches on the central magnet, with a common contraction bound $\lambda < 1$, and let τ denote the reduced stable–unstable phase increment (for the geometric application, $\tau = \log J^u F$ after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (25)$$

Proposition 6.6 (Anchored two-branch certificate). *Suppose the series in (25) converges in C^1 and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

on a central unstable interval I . Then, for all sufficiently large N ,

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.

Proof. C^1 convergence makes $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$ for all large N . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation. $\square$

Corollary 6.7 (First-term domination after acceleration). *Assume $\|Dh_i\| \leq \lambda < 1$, $\|D\tau\|_\infty \leq M$, and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

If

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

then $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$.

Proof. The derivative of the sum of all terms with $k \geq 2$ is bounded by $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1-\lambda)$. Subtract this from the first-term lower bound. $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (23); attach fixed regular connectors to the magnet; and compute Ψ'_∞ (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter Ψ_∞ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order λ for the first-term curvature separation; the remaining infinite temporal-distance tail is only $O(\lambda^2)$. This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not C^2 . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates, write the map as (F, G) with derivative $\text{diag}(\lambda, \mu)$, where $\lambda > 1$, $\mu = \lambda^{-1}$. The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (26)$$

with all derivatives evaluated at the fixed point.

Proposition 6.8 (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let $\mathcal{R} = T^2$ be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates (s, p) with $p = \sin \varphi$. Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write $S^{-1}\mathcal{R}S = (F, G)$. Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} &= F_{yy} = G_{xx} = G_{xy} = G_{yy} = 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

Proof. Place the circle centres at $(0, 0)$ and $(3, 0)$ and parametrize the first boundary near the facing point by $q(s) = (\cos s, \sin s)$. For outgoing Birkhoff data (s, p) , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at q and a target circle with centre c and radius one, put $w = q - c$, $b = w \cdot v$, and $c_0 = |w|^2 - 1$. The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates (s, p) to $(-s, -p)$ and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by S gives $G_{xyy} = -(7 - 4\sqrt{3})$; substitution in (26) gives -1 . $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength s/R and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and $\mathfrak{A}$ depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let W^u be a reference unstable leaf and let $H : \Gamma \rightarrow W^u$ be stable holonomy from a central smooth face arc. Write m_Γ for arclength and let w include the physical face velocity, coarea factor, and the smooth branch Jacobians.

Lemma 6.9 (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i) H is a $C^{1+\alpha}$ diffeomorphism onto its image and $0 < c_H \leq JH \leq C_H$;
- (ii) the SRB conditional μ^u on W^u has arclength density between $c_u > 0$ and $C_u < \infty$;
- (iii) $w \in C^\alpha(\Gamma)$ and $\|w\|_\infty \leq C_w$.

Define the signed face measure

$$\nu = H_*(w dm_\Gamma).$$

If, on every interval $J \subset W^u$, the temporal-distance phase X satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing C_1 by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with C_1 multiplied by the sum of their total-variation bounds.

Proof. The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for μ^u compare arclength and μ^u in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces. $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. This does not yet control pieces whose return word or homogeneity rank tends to infinity: their holonomy, density, and amplitude constants must be shown to grow slowly enough for Proposition 6.4 and the radial grazing sum.

6.8 A countable-IFS theorem removes the return truncation

There is a more direct route than passing through smooth finite horseshoes. Leclerc–Paukkonen–Sahlsten [13] prove power Fourier decay for Gibbs measures of finite or countable $C^{1+\alpha}$ conformal IFSs, allowing overlaps. In the countable case their assumptions are uniform contraction, bounded distortion, summable variations, and the light-tail condition

$$\sum_a \sup_{x \in I} e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} < \infty \quad \text{for some } \gamma_0 > 0. \quad (27)$$

Their Theorems 1.1–1.2 say that UNI implies QNL and that QNL gives power Fourier decay, including for Hölder amplitudes and nonstationary phases with a small polynomial loss in their norms. No finite-alphabet exhaustion or smooth extension of the branch maps is used.

This theorem is unusually well matched to the present induced problem. Let $F : I \rightarrow I$ be the unstable quotient of a full-branch Gibbs–Markov return to a central billiard magnet and let $\{h_a : I \rightarrow I\}_{a \in \mathcal{A}}$ be its inverse branches. After accelerating once if necessary, the desired dictionary is

countable-IFS input	billiard input	
h_a	homogeneous inverse return branch	
ϕ	normalized induced unstable-Jacobian potential	
uniform contraction	uniform hyperbolicity after inducing	(28)
bounded distortion	homogeneous return distortion	
light tail	return tail plus a small grazing moment	
UNI	two regular return words with separated log Jacobians.	

There is a small but important point in the first line of this dictionary. The one-sided moving-face labels must not be inserted by cutting every return branch into labelled subbranches: those pieces need not have full image and would destroy the hypothesis to which the countable-IFS theorem is applied. The correct object is a full-branch dynamical quotient with a marked amplitude fiber.

Proposition 6.10 (Marked full-branch quotient). *Let Λ be a Young return rectangle with product structure and let $\{\Lambda_a\}_{a \in \mathcal{A}}$ be a return partition such that $F|_{\Lambda_a} = T^{R_a}$ maps the*

stable subset Λ_a across Λ with the usual stable/unstable Markov property. Fix a reference unstable leaf I and let $\pi^s : \Lambda \rightarrow I$ be stable holonomy. Then

$$\bar{F} \circ \pi^s = \pi^s \circ F$$

defines a one-state full-branch expanding map: for $I_a = \pi^s(\Lambda_a)$, the map $\bar{F} : I_a \rightarrow I$ is onto and has an inverse branch $h_a : I \rightarrow I_a$.

Suppose now that the lift of a complete physical face through Λ_a is refined into one-sided marked pieces $E_{a,\ell}$, after artificial-face cancellation. On the stable quotient its action can be written

$$\nu(\varphi) = \sum_{a,\ell} \int_{J_{a,\ell}} \varphi(h_a x) w_{a,\ell}(x) d\mu^u(x), \quad (29)$$

where the $J_{a,\ell} \subset I$ are amplitude pieces and the sign is included in $w_{a,\ell}$. Thus the LPS dynamical alphabet remains $\{h_a\}_{a \in \mathcal{A}}$; the face label ℓ is a fiber mark and does not have to define a full-image branch. If

$$\sum_{a,\ell} \|w_{a,\ell}\|_{C^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty,$$

then the marked current is a summable piecewise-Hölder amplitude over the same full-branch Gibbs system.

Proof. The Markov property says that every unstable leaf in Λ_a is mapped across every stable leaf of Λ . Quotienting stable leaves therefore makes $\bar{F}|_{I_a}$ onto I ; contraction of inverse branches and bounded distortion are the usual quotient estimates. Disintegrate the coarea face measure first over return atoms and then over its one-sided physical marks. Stable holonomy transports each piece to the reference leaf and gives (29). No restriction $h_a|_{J_{a,\ell}}$ is declared to be a dynamical branch. The last assertion is the triangle inequality for the Hölder-amplitude norm. Matched artificial cuts contribute opposite weights and disappear before this disintegration. $\square$

Consequently the standard Young quotient and the moving-face bookkeeping are compatible provided the marked-amplitude sum is finite. The construction of the quotient is standard; the billiard-specific audit is precisely the last summability statement, which is supplied for the grazing tail by the bi-recovery refinement and remains to be checked for the central return alphabet.

The SRB potential part of this audit is now supported by a direct billiard result. Lemmas 5.2–5.3 of Lima–Obata–Poletti [15] prove that the collision Lebesgue/SRB measure is an equilibrium state of the geometric potential

$$\phi = -\log \|DT|_{E^u}\|, \quad P(\phi) = 0,$$

and that ϕ becomes Hölder on their countable Markov coding for finite-horizon dispersing billiards. Hence the induced potential has summable variations. This does not by itself give the one-state return tail, but it removes the former ambiguity between an MME coding and the required collision-SRB Gibbs weights.

Proposition 6.11 (Central amplitude moment). *Assume the Young return scheme has*

$$\sum_a \sup_I p_a e^{\epsilon R_a} < \infty$$

and restrict first to a compact regular homogeneity core. For a complete first moving-face defect, suppose every collision insertion has uniformly bounded C^α coarea and stable-holonomy factors. Then the marked amplitude disintegration (29) satisfies

$$\sum_{a,\ell} e^{\epsilon' R_a} \|w_{a,\ell}\|_{C^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty \quad (0 < \epsilon' < \epsilon).$$

After high-homogeneity insertions are assigned to the bi-recovery blocks of Corollary 4.4, the same conclusion holds for the entire assembled first defect.

Proof. A return word of length R_a has at most R_a possible physical insertion times. On the compact regular core each insertion contributes at most a uniform constant times the Gibbs branch weight. Hence the left-hand side is bounded by

$$C \sum_a R_a e^{\epsilon' R_a} \sup_I p_a,$$

which is finite because $Re^{\epsilon' R} \leq C_{\epsilon,\epsilon'} e^{\epsilon R}$. Artificial insertions cancel before taking total variation. The removed high-homogeneity pieces satisfy the common two-sided recovery moment and are summed by Proposition 4.1. $\square$

The following consequence records exactly what remains to be checked rather than hiding it in a finite-horseshoe approximation.

Proposition 6.12 (Countable-IFS Fourier bridge). *Suppose the central induced unstable quotient is a one-state full-branch map whose inverse branches are uniformly $C^{1+\alpha}$ contracting and have uniform bounded distortion. Suppose its normalized SRB potential has summable variations and satisfies (27). Assume there are two fixed admissible words $\mathbf{a}, \mathbf{b}$ and $c_0 > 0$ such that*

$$\inf_{x \in I} |D(\log |h'_\mathbf{a}| - \log |h'_\mathbf{b}|)(x)| \geq c_0. \quad (30)$$

Then the induced SRB Gibbs measure satisfies QNL and has power Fourier decay in the sense needed for every balanced-band phase which, after removal of the contracted coordinates, can be partitioned into uniformly finitely many pieces satisfying the phase hypotheses of the countable-IFS Fourier theorem. The same conclusion holds for the central weighted physical face slice of Lemma 6.9.

Proof. Put $Y = \log |h'_\mathbf{a}| - \log |h'_\mathbf{b}|$. The lower bound (30) and the mean-value theorem give

$$|(Y(x) - Y(y))| \geq c_0 |x - y|,$$

which is the one-state C^2 UNI condition for the inverse-branch cocycle. The countable-branch UNI theorem of Leclerc–Paukkonen–Sahlsten gives QNL, and their Fourier theorem gives the claimed power decay under (27). A finite phase partition does not change the power. Finally apply Lemma 6.9; its density and holonomy ratios only change the multiplicative constant. $\square$

The remaining two-word condition can be reduced further. The proof of Leclerc–Paukkonen–Sahlsten for their countable examples uses a finite sub-IFS and the nonlinear-IFS theorem of Algom–Rodriguez Hertz–Wang [14]: non- C^2 -linearizability produces the required pair of words. Since those finite words remain words of the countable system, no uniform search through the return alphabet is needed.

Proposition 6.13 (Finite nonlinear subsystem gives countable UNI). *Let $\{h_a\}_{a \in A}$ satisfy the contraction and distortion hypotheses of Proposition 6.12. Suppose a finite subsystem $\Phi_0 = \{h_{a_1}, \dots, h_{a_q}\}$ is uniformly contracting, C^2 , and is not C^2 conjugate to a linear IFS. Then there are two fixed words $\mathbf{a}, \mathbf{b}$ in the alphabet of Φ_0 , an integer N , and $c_0 > 0$ such that, after passing to the generation- N induced alphabet, (30) holds. In particular the entire countable IFS has UNI.*

Proof. Claim 2.1 of Algom–Rodriguez Hertz–Wang gives a generation N and two words for which

$$0 < m \leq D(\log |h'_a| - \log |h'_b|) \leq m' \quad \text{on } I.$$

The bounded-distortion error after a common prefix is exponentially small in N ; increasing N gives the positive lower bound required by UNI. This is exactly the finite-subsystem argument used in Section 7 of Leclerc–Paukkonen–Sahlsten. The same two words occur in the countable alphabet, so the countable system inherits UNI. $\square$

There is a scalar periodic-orbit certificate for the hypothesis of this proposition. It is substantially easier to check than a pointwise derivative lower bound on all of I .

Lemma 6.14 (Periodic multiplier defect). *Let h_0, h_1 be two C^2 contracting branches, let x_0, x_1 be their fixed points, and let x_{10} be the fixed point of $h_1 \circ h_0$. Define*

$$\Delta_{\text{mult}}(h_0, h_1) = \log |(h_1 \circ h_0)'(x_{10})| - \log |h'_0(x_0)| - \log |h'_1(x_1)|. \quad (31)$$

If $\Delta_{\text{mult}}(h_0, h_1) \neq 0$, then $\{h_0, h_1\}$ is not C^2 conjugate to a linear IFS and Proposition 6.13 applies.

Proof. If the pair were C^2 conjugate to affine contractions, then for some $u \in C^1$ and constants c_0, c_1 one would have

$$\log |h'_j(x)| = u(h_j x) - u(x) + c_j, \quad j = 0, 1.$$

Evaluation at x_j gives $\log |h'_j(x_j)| = c_j$. Summation along the periodic orbit of the word $(1, 0)$ gives $\log |(h_1 \circ h_0)'(x_{10})| = c_1 + c_0$, contradicting (31). $\square$

The scalar defect is useful for an explicit table, but recent homoclinic-class and QNL results remove even this calculation from the principal route.

Proposition 6.15 (Clean homoclinic horseshoe supplies a finite QNL gate). *Let p be the regular fixed point of the squared billiard map associated with the normal period-two orbit in Proposition 6.8. If its Anosov cocycle satisfies $\mathfrak{A}(p) \neq 0$, then there is a compact finite clean horseshoe Ω_p , contained in regular branches and containing p , such that in at least one time orientation its temporal distance satisfies QNL for every Hölder equilibrium state. Thus Ω_p is a finite QNL gate for the collision-SRB geometric potential.*

Proof. Lemma 5.1 of Lima–Obata–Poletti [15] says that finite-horizon dispersing billiards have a single homoclinic class at regular hyperbolic points: applying it with $x = y = p$ gives a transverse homoclinic intersection outside the full singularity orbit. The Smale–Birkhoff construction, using a finite segment of this clean homoclinic orbit, gives a compact finite horseshoe Ω_p on which the billiard map is smooth. In symplectic Birkhoff coordinates the map is area preserving; the local dynamics near Ω_p may be viewed as a smooth Axiom–A basic set.

By the contrapositive of the Anosov cocycle obstruction, together with Theorem 4.3 of Leclerc [12], in one of the two time orientations the temporal distance on Ω_p satisfies QNL for every Hölder equilibrium state. Apply this to the restriction of the collision-SRB geometric potential. $\square$

There is a regularity caveat. The stable quotient of a smooth horseshoe is in general only $C^{1+\alpha}$, precisely because the stable holonomy need not be C^2 . Therefore the finite QNL conclusion above cannot be fed directly into the C^2 non-linearizability criterion of Algom–Rodriguez Hertz–Wang. One must either verify the scalar multiplier defect (31) in a genuinely C^2 quotient, or propagate the finite QNL gate to the countable $C^{1+\alpha}$ inducing scheme. The second option is the natural billiard route.

The light-tail hypothesis has a concrete billiard reduction. If a branch a has return time R_a and grazing rank k_a , bounded distortion gives, schematically,

$$|h'_a|^{-\gamma_0} \leq C e^{c\gamma_0 R_a} (1 + k_a)^{q\gamma_0}.$$

Thus an exponential return moment and any strictly positive polynomial SRB tail in k_a imply (27) after choosing $\gamma_0 > 0$ sufficiently small. This is strictly weaker than uniform control of every rank- L smooth extension.

Lemma 6.16 (Return and grazing moments imply the LPS light tail). *Assume the normalized branch weights $p_a(x) = e^{\phi(h_a x)}$ satisfy*

$$\sum_a \sup_x p_a(x) e^{\epsilon R_a} (1 + k_a)^\delta < \infty$$

for some $\epsilon, \delta > 0$, and suppose $|h'_a|^{-1} \leq C e^{cR_a} (1 + k_a)^q$. Then (27) holds for every

$$0 < \gamma_0 < \min\{\epsilon/c, \delta/q\}.$$

Proof. Raise the inverse-derivative estimate to γ_0 and absorb the two factors into the assumed exponential and polynomial moment. $\square$

There is also a pressure-based way to obtain the required moment. The geometric-potential theory for finite-horizon dispersing billiards has a spectral gap for $-t \log J^u T$ on every compact subinterval of $(0, t_*)$, where $t_* > 1$, and its modified one-step expansion is uniform for $t \geq t_0 > 0$ [2]. Since the SRB value is $t = 1$, the LPS factor $|h'_a|^{-\gamma_0}$ merely changes the geometric exponent from 1 to $1 - \gamma_0$.

Proposition 6.17 (Geometric-pressure route to the light tail). *Let a homogeneous full-branch inducing scheme have an exponential return tail at the SRB parameter. Assume its induced distortion is uniform and that the geometric transfer estimates hold for $t \in [1 - \epsilon, 1]$. Then, after decreasing $\epsilon > 0$ if necessary, its normalized induced SRB potential satisfies (27) for every sufficiently small $\gamma_0 > 0$.*

Proof. Group branches by return time $R_a = n$. Up to the uniform distortion coboundary, the summand in (27) is the inverse unstable Jacobian to the power $1 - \gamma_0$. The geometric transfer estimate at $t = 1 - p\gamma_0$ bounds the corresponding p -moment by $C e^{c(p\gamma_0)n}$, with $c(p\gamma_0) \rightarrow 0$ as $\gamma_0 \rightarrow 0$. Hölder's inequality combines this with the SRB return tail $C e^{-an}$ to give

$$\sum_{R_a=n} \sup_x e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} \leq C e^{-an/q + c(p\gamma_0)n/p}.$$

Choose $p, q > 1$ conjugate and then γ_0 so that the exponent is negative; summing over n proves the claim. Homogeneity-strip subdivision is already included in the geometric one-step expansion. $\square$

For the usual homogeneity strips the grazing part of this lemma is expected from the collision density and the strip width; the return part is the exponential tail of the inducing scheme. It must nevertheless be verified for the particular full-branch quotient carrying the moving face. The literature theorem does not perform that billiard bookkeeping.

This route also exposes a flaw in the finite-horseshoe detour. A conservative pasting lemma [17] can locally modify or smooth a volume-preserving map, but it does not automatically extend a finite family of billiard branches to a global area-preserving Axiom–A diffeomorphism whose equilibrium state has the required collision-SRB weights. Even a successful extension can put a fractal Gibbs conditional on a saddle basic set in place of the absolutely continuous unstable conditional of the billiard SRB measure. The direct countable-IFS quotient preserves the correct measure and should therefore be the primary route.

Thus light-tail is not expected to be a new principal theorem once the standard homogeneous inducing scheme is put in the one-state form required above. The exact value $\mathfrak{A} = -1$, the clean homoclinic theorem, and Proposition 6.15 produce a finite QNL gate without naming a connector. Proposition 6.20 records the remaining gate-to-countable bridge.

6.9 Preparing the balanced-band phase

The nonstationary-phase conclusion of the LPS theorem has a genuine lower bound hypothesis on the phase derivative. Hence it cannot be invoked on the whole balanced band before the resonant small-slope set is removed. The following lemma isolates the exact interface between QNL and that theorem.

Lemma 6.18 (Frostman/finite-type good–bad decomposition). *Let μ be the Gibbs measure of Proposition 6.12. The light-tail hypothesis gives constants $s_\mu, C_\mu > 0$ such that*

$$\mu(J) \leq C_\mu |J|^{s_\mu} \quad (32)$$

for every interval J . Let $\Xi \geq 2$ and suppose the reduced balanced phase on each word piece has the form $\psi_q = \Xi \phi_q$, where the weighted sum of the C^α amplitude norms is $O(\Xi^{\epsilon_\chi})$. Assume that the normalized phases have uniform finite type K : each J_q is the union of at most N_0 intervals on which

$$\max_{1 \leq j \leq K} |\partial_x^j \phi_q(x)| \geq c_* > 0, \quad \|\phi_q\|_{C^{K+\alpha}} \leq C_*. \quad (33)$$

Then, for sufficiently small $\epsilon, \epsilon_\chi > 0$, the pieces on which $|\partial_x \psi_q| < \Xi^{1-\epsilon}$ have total weighted mass $O(\Xi^{\epsilon_\chi - \epsilon s_\mu/K})$. On every complementary component the LPS nonstationary-phase theorem applies. Consequently there is $\beta > 0$ such that

$$\sum_q \left| \int_{J_q} \chi_q(x) e^{i\psi_q(x)} d\mu(x) \right| \leq C \Xi^{-\beta}.$$

Proof. A standard one-dimensional sublevel estimate following from (33) covers $\{|\phi'_q| < \Xi^{-\epsilon}\}$ by at most $C_K N_0$ intervals of length $C \Xi^{-\epsilon/K}$. Equation (32), followed by the weighted amplitude bound, gives $O(\Xi^{\epsilon_\chi - \epsilon s_\mu/K})$ for the discarded part. On each remaining component $|\psi'_q| \geq \Xi^{1-\epsilon}$. The finite-type bound and subdivision at the endpoints of the bad intervals give the polynomial Hölder bound on $\log |\psi'_q|$ required in Theorem 1.1 of Leclerc–Paukkonen–Sahlsten. Choose $\epsilon_\chi < \epsilon s_\mu/K$ and then take all polynomial losses below the thresholds of that theorem. Add the good and bad contributions. $\square$

This corrects a tempting but false shortcut: LPS QNL concerns fluctuations of the log-derivative cocycle and does not, by itself, imply a small-value bound for an arbitrary phase slope. Small-slope pieces are instead controlled by the Frostman estimate and finite type. After the contracted coordinates are removed, the central radial problem is two-dimensional; the preferred check is $K = 2$, equivalently a uniform nonzero Wronskian

$$|\det(\Gamma'_{m,n}(t), \Gamma''_{m,n}(t))| \geq c_* > 0 \quad (34)$$

for the normalized reduced plane curve. Lemma 6.3 supplies the local curvature seed, but its uniform persistence through the chosen past/future return words is a finite geometric calculation still to be written.

The reduced curvature has an exact cocycle form. This identifies the right way to combine the finite QNL gate with the phase decomposition.

Lemma 6.19 (Wronskian/log-slope identity). *Let X, Y be monotone C^2 functions and put $\Gamma = (X, Y)$. Then*

$$\det(\Gamma', \Gamma'') = X'Y' D \log \left| \frac{Y'}{X'} \right|. \quad (35)$$

Consequently, if the normalized branch coordinates satisfy $c \leq |X'|, |Y'| \leq C$ and

$$|D(\log |Y'| - \log |X'|)| \geq c_0,$$

then the reduced curve has uniform type $K = 2$.

Proof. Expand $X'Y'' - Y'X'' = X'Y'(Y''/Y' - X''/X')$. □

Proposition 6.20 (QNL-gated triple inducing). *Let a one-state countable Gibbs–Markov quotient contain the finite clean QNL gate Ω_p of Proposition 6.15. Induce again from the central magnet to the first return which has passed through a fixed entrance block, a QNL gate block, and a fixed exit block. Assume:*

- (G1) *the gate-hitting time has an exponential tail;*
- (G2) *conditioned on every entrance/exit context, the collision-SRB conditional temporal-distance fluctuation on the gate satisfies the uniform small-ball estimate $C\sigma^\Gamma$;*
- (G3) *after hyperbolic normalization, the reduced coordinate maps X, Y have derivatives bounded above and below and bounded $C^{1+\alpha}$ distortion uniformly over the induced alphabet.*

Then the triple-induced countable system satisfies QNL. Moreover, for every $r \in (0, 1)$ the total Gibbs weight of branch pairs on which

$$|D(\log |Y'| - \log |X'|)| < r$$

at the resolving scale is $O(r^\Gamma)$ for some $\Gamma > 0$. On the complement, Lemma 6.19 and Lemma 6.18 give the balanced-band Fourier bound.

Proof. Disintegrate an induced branch into its entrance context, gate block, and exit context. The logarithmic derivative is additive under composition. Once the contexts are fixed, their contribution translates the gate fluctuation; the small-interval estimate in (G2) is uniform under translations. Integrating the contexts proves QNL for the induced system. Condition (G1) preserves the light tail when lifting from the triple-induced clock to the original return clock.

At a resolving interval, bounded $C^{1+\alpha}$ distortion turns a small log-slope derivative into a small gate fluctuation on a comparable subinterval; the QNL small-ball estimate therefore gives $O(r^\Gamma)$ after a harmless change of exponent. On the complementary branch pairs, (35) gives type two, with constants allowed to lose a fixed power of r^{-1} . Choose $r = \Xi^{-\delta}$ with $\delta > 0$ small. The bad branch weight and the polynomial loss on the good branches are both absorbed by the power saving in Lemma 6.18. $\square$

Condition (G2), rather than the existence of a nonlinear periodic orbit, is now the precise central go/no-go. It is a local-product non-concentration theorem on a fixed clean gate. Bertozzi's triple-inducing construction is the closest published template, but it treats a differentiable UNI roof rather than a billiard temporal-distance QNL gate.

Condition (G2) is not a formal consequence of the finite horseshoe QNL in Proposition 6.15. The horseshoe equilibrium conditional is supported on a Cantor set, whereas the collision-SRB unstable conditional in the magnet is absolutely continuous; the two conditionals are not related by a bounded density. What is required is a singular extension of Leclerc's template/non-concentration proof directly to the countable first-return system, using SRB local product structure, the clean homoclinic recurrence, and the exponential return tail. This support mismatch is the last genuinely new theorem in the route.

By contrast, (G3) is standard. If $h : I \rightarrow I_h$ is an inverse return branch, affinely normalize I_h to unit length. Uniform bounded distortion gives upper and lower bounds for the derivative of the normalized branch, and the $C^{1+\alpha}$ distortion estimate gives a uniform Hölder bound for its log derivative. Apply this separately to the normalized past-stable and future-unstable coordinates. Thus only (G2), not (G3), remains open.

7 An induced QNL/Fourier program

The most economical current route is the following.

- (F1) Construct an exponential return scheme on a regular magnet and retain both one-sided codes. Quotient stable leaves to obtain a one-state countable full-branch $C^{1+\alpha}$ IFS with collision-SRB Gibbs weights. Keep one-sided physical labels in the amplitude fiber of Proposition 6.10, rather than cutting the dynamical branches. Verify uniform contraction, bounded distortion and the light-tail moment in Lemma 6.16. An exact generalized-baker skew product and a smooth finite-horseshoe extension are unnecessary.
- (F2) Lift each complete physical face to finitely or countably many smooth graph slices of the return rectangles. Artificial cuts cancel before the lift, and the two marginals are removed before estimating any branch.
- (F3) Use Lemma 6.2 to reduce to $m \asymp n$. Proposition 6.15 supplies a finite QNL gate in one time orientation; triple induce through it as in Proposition 6.20. In adapted stable-unstable product coordinates, remove the contracted coordinates and prove uniform nonlinearity/non-concentration for the dominant two-coordinate phase induced by

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m}\gamma_b(t) + \zeta \cdot T^n F_b \gamma_b(t). \quad (36)$$

The associated temporal distance, expressed through stable holonomy, satisfies UNI and the weighted small-value estimate by the finite nonlinear subsystem. A direct full-product Fourier proof would instead require a uniform finite-type condition through at least fourth order and is not the preferred route.

- (F4) Apply Proposition 6.12 directly to the countable induced SRB Gibbs measure. Partition the reduced balanced phase by Lemma 6.18; the Frostman estimate controls small-slope pieces once the reduced finite-type test (33) is verified. Use L^2 flattening only if the remaining branchwise cancellation is not covered by the countable-IFS Fourier theorem.
- (F5) Lift the return-time estimate to collision time using Corollary 4.4 and Proposition 4.1. The preferred target is $R_b^\pm = O(1+b)$ with refined total variation $O(M_b)$; then the generic $M_b \lesssim 2^{-b}$ bound already gives

$$\sum_b M_b(1 + R_b^-)(1 + R_b^+) < \infty.$$

If instead Fourier analysis is applied directly on each unrecovered grazing curve, one needs the stronger, nonpreferred margin

$$\sum_{b \geq 0} 2^{-b} C_b < \infty. \quad (37)$$

A polynomial deterioration $C_b \lesssim 2^{\kappa b}$ is acceptable only for $\kappa < 1$. Large grazing expansion can help the oscillatory estimate, but its distortion cost must be tracked in the same inequality. For the Markov-normalized first radial one-face graph, the fixed-atlas flux calculation is sharper: its grouped singular coefficient is $O(2^{-(3-\epsilon_*)b})$, while the weaker $O(2^{-(1-\epsilon_*)b})$ term is a regular two-dimensional interior/Jacobian source. If the oscillatory factorization preserves that separation, the graph-current margin improves to

$$\sum_{b \geq 0} 2^{-(3-\epsilon_*)b} C_b < \infty,$$

allowing $\kappa < 3 - \epsilon_*$. The recovery-window route avoids this exponential constant altogether. Simultaneous properness is supplied by the two marginal growth trees for the clean radial first defect.

- (F6) Establish the same estimates uniformly for the assembled difference quotients

$$s^{-1}(P_s - P_0),$$

by verifying (U1)–(U4) of Proposition 7.4, and then apply dominated convergence.

The recent symbolic construction of Climenhaga–Day gives a concrete finite-alphabet two-sided coding and product structure for every finite-horizon Sinai billiard map. It is useful as a geometric skeleton, but it constructs the measure of maximal entropy. CM2 concerns the collision SRB/flux measure, so its Gibbs weights, holonomies, and parameter-uniform distortion still have to be built into the induced scheme. This measure mismatch cannot be hidden.

A possible but now secondary route is a finite-horseshoe approximation. Truncate the return and homogeneity ranks at L and retain a fixed regular periodic word witnessing nonlinearity. Even if the resulting finite branch core can be realized as a smooth area-preserving Axiom–A basic set, one must still match its equilibrium state to the collision SRB weights and prove uniformly

$$\mu\{\text{an orbit segment sees rank} > L\} \leq CNe^{-aL}, \quad (38)$$

$$(\mu_L \otimes \mu_L)\{|\Delta_L| \leq \sigma\} \leq Ce^{cL}\sigma^\Gamma, \quad (39)$$

and the analogous estimate for the weighted face slice. A nonzero periodic-point cocycle in a fixed central branch is a plausible uniform QNL witness, but the published smooth theorem does not state uniformity under this singular exhaustion. Proposition 6.12 bypasses (39) entirely when the induced quotient satisfies its assumptions. The finite truncation is retained only as a fallback.

The standard-pair equidistribution theorem of Bálint–Nándori–Szász–Tóth proves that one-dimensional singular initial measures can mix exponentially after a suitable smoothing argument. It supplies the future-time benchmark, not the product estimate: smoothing in the future-stable direction makes the past factor in (17) rougher, and reversing the smoothing exchanges the problem rather than solving it.

Finally, the large-essential-spectrum result of Butterley–Canestrari–Castorrini warns against replacing this dynamic program by a globally stronger Banach norm which controls every discontinuity trace. For piecewise two-dimensional systems such a norm can retain large essential spectrum. This is consistent with the local collar counterexample and favors an orbit-dependent oscillatory estimate.

7.1 Execution audit of the seven links

The seven links now have the following status.

- (1) *Return coding and marked amplitudes: closed from standard data.* Exponential Young structures and local product rectangles are standard [16] for finite-horizon dispersing billiards. The countable-IFS theorem means that no smooth finite truncation is required. What must be stable quotient is a one-state full-branch map with the standard branch distortion and exponential return moment. Proposition 6.10 shows that the moving face and its one-sided labels belong to a marked amplitude fiber and therefore do not destroy the full-image property. Proposition 6.11 proves the central marked-amplitude summability; high homogeneity is assigned to the recovery tree.
- (2) *SRB weights and light tail: closed from geometric pressure.* Markov normalization identifies the collision invariant density and the induced Gibbs potential is the unstable-Jacobian potential. The direct countable coding of Lima–Obata–Poletti makes this geometric potential Hölder and identifies the collision SRB equilibrium state. Proposition 6.17 derives the LPS moment from the known interval of geometric potentials and an exponential return tail.
- (3) *Face lift: locally passed.* Artificial faces cancel before the coarea current is formed, the central radial tangency graph has nonzero coarea derivative, and Lemma 6.3 gives an explicit non-affinity certificate. Lemma 6.9 transfers central-block QNL to its physical weighted face measure. A global bi-return partition of the face remains to be written.
- (4) *Quantitative nonlinearity: finite gate closed, propagation explicit.* Proposition 6.8 gives an exact nonzero Anosov cocycle at a fixed normal two-cycle. The countable alphabet and return-rank exhaustion are handled by Proposition 6.12. Proposition 6.15 uses the single-homoclinic-class theorem and Leclerc to produce a finite QNL gate. Because the stable quotient is only $C^{1+\alpha}$, one cannot automatically invoke the C^2 nonlinear-IFS criterion. The exact remaining propagation condition is (G2) of Proposition 6.20.

- (5) *Fourier conversion: reduced to gated curvature.* The countable-IFS theorem supplies power decay directly after the reduced balanced phase is partitioned into nonstationary pieces. Lemma 6.18 closes the good/bad estimate from the LPS Frostman bound once the reduced finite-type condition (33) is verified. Lemma 6.19 identifies the curvature with the log-slope cocycle, while Proposition 6.20 controls the statistically degenerate branch pairs. Proposition 6.1 then gives the product rate; the frequency scale is the maximum of the two hyperbolic scales, costing only one half of the Fourier exponent.
- (6) *Tower and grazing lift: closed for the clean radial first defect.* Proposition 6.4 handles exponential return tails. Proposition 4.1 shows that $M_b = O(2^{-b})$ and $R_b^\pm = O(b)$ suffice, and Corollary 4.4 supplies the required two-sided growth-tree refinement. The stronger exponent $3 - \epsilon_*$ is reserve margin.
- (7) *Parameter difference quotient: closed by the master refinement.* Proposition 7.3 constructs one stratified refinement in parameter-state space and proves (U1) and (U4). Proposition 7.4 then gives dominated convergence.

Accordingly the program does not yet prove CM2. The local nonlinearity is no longer conjectural: it has the exact witness (26), the finite QNL gate is constructed, and the parameter difference quotient is closed by a master coarea refinement. The positive theorem is now reduced to one central gate audit: verify (G2) in Proposition 6.20. Through the exact Wronskian identity this simultaneously supplies countable QNL and the normalized finite-type estimate. The grazing-rank problem is separated and summed by Corollary 4.4.

7.2 A quantitative smoothing test

There is one falsifiable way to obtain (16). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (40)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (41)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad qquad \Lambda_+ e^{-a\alpha} < 1. \quad (42)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (41) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (41). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 7.1 (Uniform difference-quotient closure). *Suppose either (14)–(15), or the shifted estimate (11) and the summability condition on the right of (12), hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$. If each block converges weakly to $K_{V,b}$, then the parameter difference quotient of the prescribed susceptibility has a summable majorant and converges term by term to the double series generated by K_V .*

Proof. In the first case apply Theorem 3.2; in the second apply Proposition 4.1. Either hypothesis gives an integrable majorant after summing over b, m, n . Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

For the moving table, the uniform hypothesis in this proposition can itself be reduced to four checkable pieces.

Lemma 7.2 (Parameter derivatives of contracting return words). *Let $h_{j,s}$ be the inverse one-step branches in a fixed compact regular atlas and assume, uniformly in j and $|s| < s_0$,*

$$\|Dh_{j,s}\|_\infty \leq \lambda < 1, \quad \|\partial_s h_{j,s}\|_{C^1} + \|\partial_s \log |h'_{j,s}|\|_\infty \leq A.$$

For a clean word $a = (a_1, \dots, a_R)$ let $h_{a,s} = h_{a_1,s} \circ \dots \circ h_{a_R,s}$. Then

$$\|\partial_s h_{a,s}\|_\infty \leq \frac{A}{1-\lambda}, \quad \|\partial_s \log |h'_{a,s}|\|_\infty \leq C(1+R). \quad (43)$$

The same linear-in- R bound holds for products of uniformly C^1 Jacobian, stable-holonomy, and coarea factors. Hence an exponential return moment dominates the difference quotients of all clean marked amplitudes and their return tail.

Proof. Differentiate the composition. The contribution created at position k is multiplied on the left by at least $k-1$ contracting derivatives, so its C^0 norm is bounded by $A\lambda^{k-1}$; summation gives the first bound. The logarithmic derivative of a composition is a sum of R one-step logarithmic derivatives. Differentiating it gives a uniformly bounded direct term at every position and orbit-displacement terms controlled by the first bound and uniform C^2 distortion. This gives $C(1+R)$. The product statement follows after taking logarithms. Finally $R^p e^{-\epsilon R}$ is summable for every fixed p , so the exponential return moment absorbs all these factors. $\square$

This lemma shows that long return words do not create an exponential parameter-derivative loss on the inverse-branch side. The remaining issue is not the clean word calculus but the common refinement of words that approach or cross a moving singularity; those pieces must be charged to the already constructed face/recovery tree rather than differentiated as clean branches.

That singular refinement is most naturally made before fixing the parameter.

Proposition 7.3 (Parameter-state master refinement). *Let $G_{a,k}(s, x)$ be the physical branch inequalities encountered at time $k < R_a$ by return word a , written in the transported collision space. Assume the compact family has uniform simple-face transversality, the central marked moment of Proposition 6.11, and the two-sided recovery moment of Corollary 4.4. Partition $(-s_0, s_0) \times M$ by the signs of all $G_{a,k}$, retaining the physical one-sided label on their zero sets and cancelling artificial cuts. Then the s -slices form a common measurable refinement of all return partitions and the assembled first difference quotient satisfies*

$$\frac{\mathbf{1}_{\{G_{a,k}(s, \cdot) > 0\}} - \mathbf{1}_{\{G_{a,k}(0, \cdot) > 0\}}}{s} = \int_0^1 \delta(G_{a,k}(\theta s, \cdot)) \partial_s G_{a,k}(\theta s, \cdot) d\theta \quad (44)$$

in distributions. Its clean return pieces and recovered grazing pieces obey the uniform majorant (45). In particular (U1) and (U4) of Proposition 7.4 hold.

Proof. For a test function, the fundamental theorem of calculus applied to $H(G_{a,k}(s, x))$ gives (44). Coarea changes the delta term into the one-sided face density

$$\frac{\partial_s G_{a,k}}{|\nabla_x G_{a,k}|} d\sigma_{a,k}.$$

Uniform physical transversality bounds this density on the compact regular core. A word of length R_a has at most R_a insertion hypersurfaces, so their clean total variation is summable by Proposition 6.11. At first order, intersections of two clean faces have codimension two and carry no additional measure; artificial faces cancel before total variation is taken.

For grazing pieces, use the same formula in the square-root blown-up charts. The uniform renormalized physical block bound and the common past/future growth-tree refinement give

$$\sum_{b,j} M_{b,j}(1 + R_{b,j}^-)(1 + R_{b,j}^+) < \infty$$

uniformly for every intermediate parameter θs . Proposition 4.1 therefore gives (45). Regular density/Jacobian terms are handled by the ordinary fundamental theorem of calculus and Lemma 7.2. Finally, the master hypersurfaces converge on every fixed label as $s \rightarrow 0$; the same summable majorant gives weak convergence by dominated convergence. $\square$

This construction is why a differentiable parameter family of Young towers is unnecessary. The dynamical return alphabet is fixed on clean master cells, while every atom which changes its itinerary is represented by the physical coarea current that the response formula already contains.

Proposition 7.4 (Common-refinement majorant for moving quotients). *Let $|s| < s_0$ and transport the billiard maps to the base collision space. Assume:*

- (U1) *the clean return atoms admit a common marked refinement on which $h_{a,s}$, the stable holonomy, the coarea weight, and the reduced phase are C^1 in s ; the discarded atoms have a uniform exponential return tail;*
- (U2) *contraction, distortion, the LPS light-tail moment, and the piecewise-Hölder marked-amplitude sum of Proposition 6.10 are uniform in s ;*
- (U3) *the finite clean homoclinic subsystem of Proposition 6.15 persists, and (G2) of Proposition 6.20 holds uniformly in s ;*
- (U4) *the assembled difference-quotient blocks have a common recovery majorant*

$$|B_{s,b,j}(m,n)| \leq CM_{b,j}\tau^{(m-R_{b,j}^-)_+}\vartheta^{(n-R_{b,j}^+)_+}, \quad \sum_{b,j} M_{b,j}(1 + R_{b,j}^-)(1 + R_{b,j}^+) < \infty, \quad (45)$$

with the sum independent of s , and every fixed block converges weakly as $s \rightarrow 0$.

Then the balanced-band LPS bound, the two off-diagonal coupling bounds, and (45) give a common summable majorant for $s^{-1}(P_s - P_0)$. Consequently the susceptibility difference quotient converges to the CM2 double series.

Proof. By (U3), hyperbolic continuation, and compactness in s , Proposition 6.20 gives uniform QNL and the required finite-type decomposition. Conditions (U1)–(U2) make the Gibbs, amplitude, and light-tail constants in the LPS theorem uniform. Lemma 6.18 therefore gives the same power bound on the balanced band for every s . The forward and time-reversed standard-family estimates handle the two off-diagonal regions as in Lemma 6.2. Finally sum (45) by Proposition 4.1. The right-hand side is independent of s ; weak convergence on fixed common-refinement atoms and dominated convergence finish the proof. $\square$

The published moving-scatterer memory-loss theorem alone does not construct the C^1 common inducing refinement. Proposition 7.3 supplies it by working in parameter-state space and assigning every itinerary change to its physical coarea face. Thus no differentiable family of Young towers is required.

8 Next falsifiable tests

The central calculation is now a single explicit gate audit: verify (G2) of Proposition 6.20. The exact identity (35) then turns the gated log-derivative non-concentration into the normalized finite-type estimate, while Proposition 7.3 supplies (U1)–(U4). The finite QNL gate is supplied by Proposition 6.15, while the quotient weights and marked amplitudes are supplied by Propositions 6.10 and 6.11. This gate audit implies the local estimate

$$\left| \int_I g(T^{-m}\gamma(t)) f(T^n F\gamma(t)) w(t) dt \right| \leq C\tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}. \quad (46)$$

using Propositions 6.15, 6.20, 6.12, and 7.4, together with the weighted-transfer lemma. The remaining calculation is a local-product/distortion comparison on one fixed clean gate, not a return-rank exhaustion problem.

The high-radial-block check is now an audit rather than a new mixing theorem: write the common past/future growth-tree refinement explicitly and match its weights to the fixed-atlas physical block bound used in Corollary 4.4. High grazing ranks then contribute only

$$\sum_b M_b(1+b)^2 < \infty,$$

and no blockwise Fourier constant is needed. A failure of this bookkeeping would invalidate the corollary, but it would not reopen the central QNL calculation.

In parallel, one may still test the stronger mixed-cylinder condition (16). Success would imply CM2 immediately, but failure would only show that the graph current is non-nuclear; it would not refute (46).

The current status is therefore precise: future recovery is technically available, complete centering is algebraic, the prescribed source is already independent and is not a coboundary, and a fixed circular normal cycle gives an exact nonzero QNL seed. The newest countable-IFS theorem removes the smooth singular exhaustion from the principal path. The label/full-branch conflict and the abstract good/bad phase estimate are now resolved by Proposition 6.10 and Lemma 6.18. The unresolved billiard work is to verify the conditional-density comparison and normalized coordinate distortion in (G2). Existing projective cones and one-sided coupling do not perform this gate comparison. This is the best falsifiable route found in the present audit.

References

- [1] M. Demers and C. Liverani, *Projective cones for sequential dispersing billiards*, Commun. Math. Phys. 401 (2023), arXiv:2104.06947.
- [2] M. Demers and C. Liverani, *Recent progress in the application of transfer operators to dispersing billiards*, arXiv:2606.10155 (2026).
- [3] M. Stenlund, L.-S. Young and H.-K. Zhang, *Dispersing billiards with moving scatterers*, arXiv:1210.0011.

- [4] J. Leppänen and M. Stenlund, *A note on the finite-dimensional distributions of dispersing billiard processes*, arXiv:1702.01461.
- [5] C. Wormell, *Conditional mixing in deterministic chaos*, arXiv:2206.09291; and *On convergence of linear response formulae in piecewise hyperbolic maps*, arXiv:2206.09292.
- [6] S. Baker and A. Banaji, *Polynomial Fourier decay for fractal measures and their pushforwards*, Math. Ann. 392 (2025), 209–261, arXiv:2401.01241.
- [7] S. Baker, O. Khalil and T. Sahlsten, *Fourier decay from L^2 -flattening*, arXiv:2407.16699.
- [8] V. Climenhaga and J. Day, *Every finite horizon Sinai billiard map has a unique measure of maximal entropy*, arXiv:2604.25881.
- [9] P. Bálint, P. Nándori, D. Szász and I. P. Tóth, *Equidistribution for standard pairs in planar dispersing billiard flows*, Ann. Henri Poincaré 19 (2018), arXiv:1707.03068.
- [10] O. Butterley, G. Canestrari and R. Castorrini, *Discontinuities cause essential spectrum on surfaces*, Ann. Henri Poincaré (2024), arXiv:2306.00484.
- [11] G. Leclerc, *Fourier decay of equilibrium states for bunched attractors*, arXiv:2301.10623.
- [12] G. Leclerc, *Fourier decay of equilibrium states and the Fibonacci Hamiltonian*, arXiv:2507.23731v2 (2026).
- [13] G. Leclerc, S. Paukkonen and T. Sahlsten, *Fourier decay in parabolic $C^{1+\alpha}$ systems with overlaps*, arXiv:2505.15468v3 (2025).
- [14] A. Algom, F. Rodriguez Hertz and Z. Wang, *Polynomial Fourier decay and a cocycle version of Dolgopyat’s method for self conformal measures*, arXiv:2306.01275v2 (2024).
- [15] Y. Lima, D. Obata and M. Poletti, *Measures of maximal entropy for non-uniformly hyperbolic maps*, arXiv:2405.04676v2 (2025).
- [16] N. Chernov, *Decay of correlations and dispersing billiards*, J. Stat. Phys. 94 (1999), 513–556.
- [17] P. Teixeira, *On the conservative pasting lemma*, Ergodic Theory Dynam. Systems 40 (2020), 1402–1440, arXiv:1611.01694.
- [18] R. Rühr, *Pressure inequalities for Gibbs measures of countable Markov shifts*, arXiv:2012.13226v3.
- [19] N. Bertozzi, *Exponential mixing for hyperbolic flows on non-compact spaces*, arXiv:2603.11852v2 (2026).
- [20] D. Galli, *A cohomological approach to Ruelle–Pollicott resonances and speed of mixing of Anosov diffeomorphisms*, arXiv:2405.17045.